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Feasibility study for diagnosis of stomach adenoma and cancer using IR spectroscopy

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Abstract

The feasibility of infrared (IR) spectroscopy as a biomedical analysis tool for the diagnosis of stomach malignancy including adenoma and cancer has been studied using unstained biopsy samples. Biopsy samples were acquired from 11 subjects. IR spectra were collected for these samples using a microscope (aperture: 25 μm × 25 μm). The samples were stained again and the spots where the IR spectra were collected were re-examined by a pathologist to ensure the spectra represented the correct diagnostic information. The spectral features were compared among the averaged spectra of normal and malignant tissues. The spectral contrasts could be correlated to the differences in the molecular structure of the membrane lipids of the two tissue types as well as the variation in their glycogen contents. However, the spectral features between the adenoma and cancer tissues could not be distinguished. Initially we used principal component analysis (PCA) to examine the degree of separation between tissue types. Soft independent modeling of class analogies (SIMCA) was employed to evaluate the prediction accuracy of IR spectroscopy for the diagnosis of stomach adenoma and cancer. The prediction accuracies for normal, adenoma and cancer tissues were 77%, 30% and 87%, respectively, using SIMCA. IR microscopy successfully differentiated normal and malignant tissues. However, a more sophisticated algorithm will be required in order to effectively extract relevant information for the differentiation between stomach adenoma and cancer.
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1. Introduction

Vibrational spectroscopies such as infrared and Raman microscopy have been studied for the diagnosis of diverse diseases, especially for various cancers [1–12]. The motivation to use spectroscopic methods for biomedical diagnoses is based on the fact that these analytical techniques can provide rich pathological information related to the difference in chemical structure between normal and malignant tissues; therefore, the accuracy as well as the objectivity of diagnoses can be improved when combined with conventional visual inspection of a stained sample by a pathologist. Vibrational spectroscopy has been studied for the diagnosis of different cancers such basal cell carcinoma (BCC), colon and breast cancer [13–19]; however, there are few detailed reports on the spectroscopic diagnosis of stomach cancer or adenoma. Recently, Fujioka et al. reported the differentiation between malignant and normal human gastric tissues using infrared (IR) spectroscopy [20,21]. These workers manually prepared samples that were 1 mm^3 in size by cutting portions from biopsy tissues. They pressed them to achieve sample thicknesses of 10 μm and employed IR transmission measurements to collect data. This sampling method cannot be used for mapping or imaging purposes for early diagnosis because the original pathological identity could be lost due to squeezing of the tissue samples. In such a case, the acquired IR spectrum would be the averaged spectrum of the whole section of the biopsy since a large amount of sample was prepared for the measurement.

Additionally, Yu et al. attempted to distinguish malignant from normal stomach tissues using ATR-IR spectroscopy [22].

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They scanned sample areas that were 1–2 cm in diameter. The spectral features from this approach were representative of the entire sample area in each case. Though useful information can be collected in this manner, it would be beneficial to be able to collect pinpoint data that correlated to localized information.

The major goal of this work was to present the feasibility of IR spectroscopy to directly measure real biopsy samples without the need for staining. Ultimately, the use of this technique for the diagnosis of stomach malignancy including adenoma and cancer would be desirable. Gastric adenomas are considered as premalignant lesions (from adenoma through adenoma–carcinoma sequence) [23,24]. Since IR spectroscopic imaging can be performed using a small aperture, early diagnosis is feasible by acquiring information from tiny malignant spots. The speed of diagnoses can be greatly enhanced because the IR image of a sample (without staining) can be obtained in a few minutes when an array detector is used.

Biopsy samples from eleven different subjects were used for this study (four adenoma and seven cancer subjects). Two sections were prepared from each subject: one section for H&E stain and the other for IR measurement. Using an IR microscope (aperture: 25 μm × 25 μm), IR spectra of unstained normal, adenoma and cancer tissues were collected. The portions of the tissues set aside for staining were used to reference the positions for IR data collection. To confirm that the IR spectra were representative of the pathology information, the spots used for data collection were confirmed by staining and examination by a pathologist.

We used principal component analysis (PCA) as well as soft independent modeling of class analogies (SIMCA) [25,26] as chemometric methods to discriminate normal, adenoma and cancer tissues. Reasonable spectral differences were observed between the spectra collected from normal and malignant tissues. However, the spectral features between stomach adenoma and cancer were not easily discriminated. This conclusion came from the fact that the score clusters of the adenoma and cancer tissues were significantly overlapped. Using SIMCA, the prediction accuracies for the normal and cancer tissues were quite acceptable (77% and 87%, respectively). The prediction accuracy for adenoma could perhaps be improved by the development and application of more sophisticated chemometric algorithms.

2. Experimental

2.1. IR spectra and data processing

IR spectra were collected using an FT-IR spectrometer equipped with a microscope (SensIR Technologies, Danbury, CT). The aperture size was 25 μm × 25 μm and an MCT detector was used. Malignant stomach tissues from eleven different patients were acquired from Guro Medical Center of Korea University, Seoul, Korea. All the samples were obtained according to the proper ethical guidelines with permission obtained from all subjects. The samples were cut (10 μm in thickness) and transferred onto an e-glass (coated with IR reflective material) for IR measurements. Because of this, the collected IR data were the result of double transmission measurements. IR spectra were acquired by accurately positioning the incident beam on different areas of the samples based on diagnostic information from the pathology testing (H&E stained sample). IR spectra were obtained by collecting 32 scans at a resolution of 4 cm^{-1}. Overall, 926 spectra were collected from 11 biopsy samples. A detailed description of the collected spectra is summarized in Table 1. All of the IR spectra were collected at room temperature (20 °C). The spots used for IR analysis were carefully selected and most of the data were collected from mucous cells in the mucosa layer rather than the lumen or outside of the gland, since adenoma or cancer can originate from alteration of the mucous cells.

Table 1
The description of the whole data set

	No. of spectra collected		
	Normal	Adenoma	Cancer
Subject 1	10	10	0
Subject 2	32	29	0
Subject 3	45	0	41
Subject 4	0	0	60
Subject 5	55	0	65
Subject 6	0	0	100
Subject 7	40	0	40
Subject 8	39	0	25
Subject 9	84	89	0
Subject 10	43	48	0
Subject 11	0	0	71
Total	348	176	402

PCA, SIMCA and all other pre-processing methods were accomplished using Matlab Version 7.0 (The MathWorks Inc., MA, USA).

3. Results and discussion

3.1. Sample image and spectral features

Fig. 1(a) shows the image of the stained biopsy sample containing normal and adenoma tissues. The targeted areas for IR spectral collection were established based on a pathologist's decision, and are shown as dotted line squares in the figure. After the targeted areas were established, we collected IR spectra using a parallel unstained biopsy sample that was consecutively sectioned during sample preparation. Fig. 1(b) shows the magnified image of the unstained sample that corresponds to the two targeted areas designated in Fig. 1(a). The circles with numbers are the actual spots where the IR measurements were made. In the course of spectral collection, the most difficult step was finding the correct corresponding spot in the unstained sample based on the stained reference image since these two visual images were quite different from each other. Furthermore, the two-dimensional images of the two biopsy samples could be slightly different from each other, even though these were sectioned consecutively. Hence, the most critical concern was to acquire IR spectra to correctly represent the pathological information. Therefore, we decided

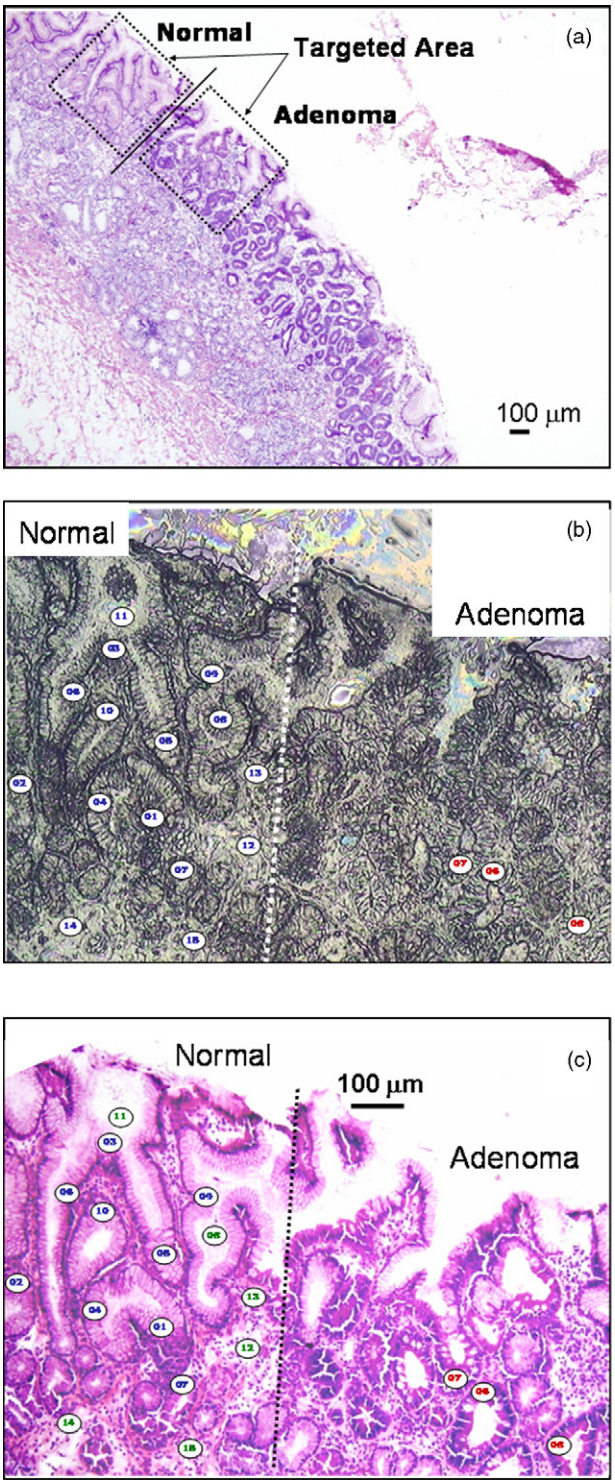

Fig. 1. The stained image of the biopsy sample contains normal and adenoma tissues (a), magnified image of the unstained sample that corresponds to two targeted areas (b) and re-stained image for the same sample after IR collection (c).

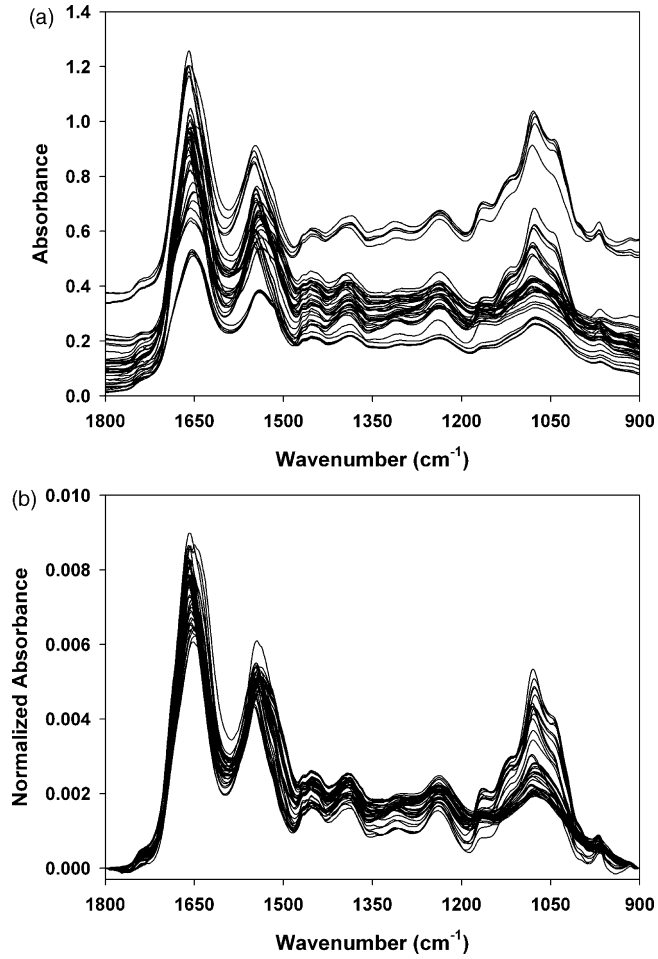

Fig. 2. Twenty four raw IR spectra from eight subjects (normal tissues) (a) and the corresponding pre-processed spectra using two point baseline correction at 1800 and 900 cm^{-1}, and normalization by the area under the 1800–900 cm^{-1} range (b). Three spectra were randomly selected from each subject.

to stain the sample used to collect IR spectra again and re-examine the spots where data were collected to see whether these were properly assigned with correct pathological information after spectral collection. In this way, we improved the reliability of the IR spectra by correctly representing pathological identity. The resulting re-stained image is shown in Fig. 1(c). By comparing with Fig. 1(b) and (c), we could confirm that the collected IR spectra contained the correct pathological identity.

The thickness at the measurement spot might not be constant over an entire tissue sample. Therefore, the absorption magnitude of the collected spectra could change due to thickness variation rather than concentration or compositional variation. This kind of unwanted spectral variation should be minimized or removed for accurate biomedical diagnosis. Ideally, the IR spectra should contain spectral variation solely based on compositional or structural changes without the influence from sampling or instrumental changes. Fig. 2(a) shows 40 IR spectra from normal tissues obtained from eight subjects. Five spectra were randomly selected from each sample. As shown, the most noticeable feature of the data shown in Fig. 2(a) is the baseline variation among the subjects. It should be noted that the baseline variation in a given subject is significantly smaller than the variation among the various subjects. The baseline variation mostly originates from the slight difference in distance between IR object lens and the glass slide with each measurement. Therefore, the baselines of the spectra were initially corrected and normalized by intensity at both 1800 and 900 cm^{-1}, and also

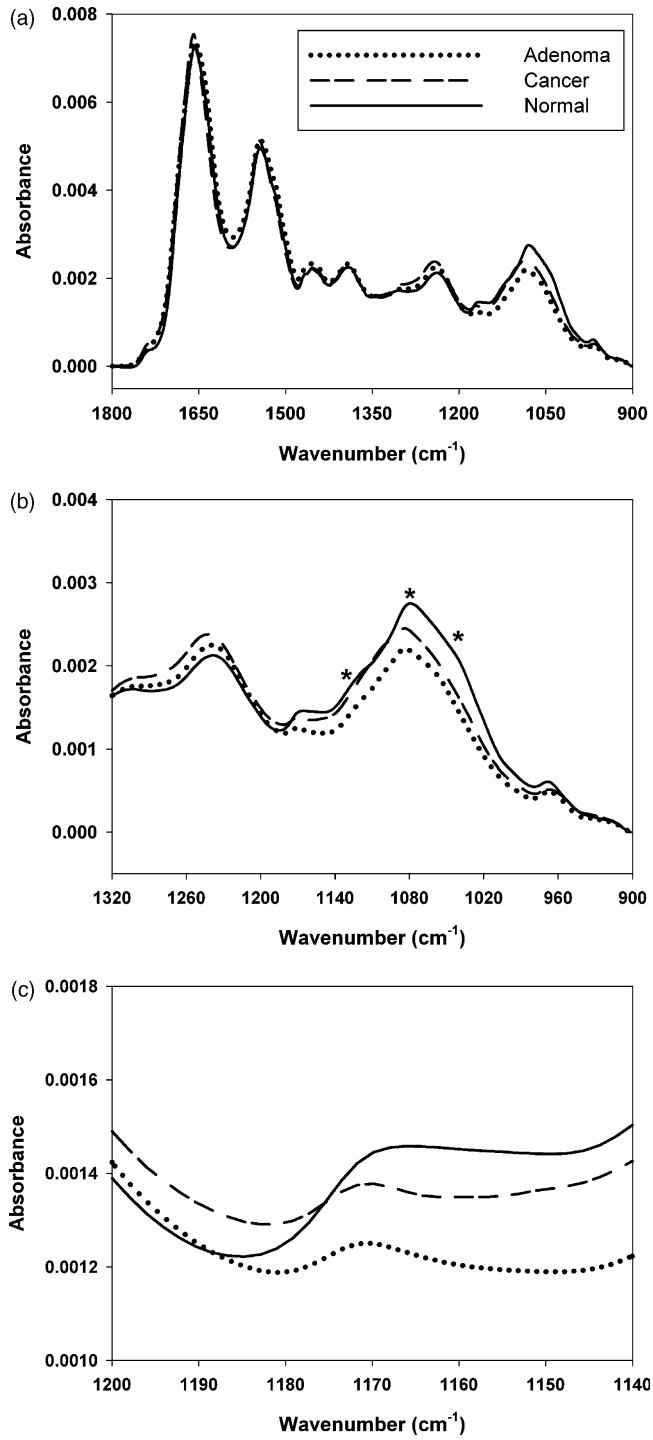

Fig. 3. Average IR spectra of normal, adenoma and cancer tissues from all subjects.

normalized by the area under the 1800–900 cm^{-1} range. By using area normalization, the unwanted spectral variations from different sample thicknesses at the measurement spots were reduced. The resulting treated spectra are shown in Fig. 2(b).

Fig. 3(a) shows the averaged IR spectra of all normal, adenoma and cancer tissues. As shown, the overall spectral features are generally similar. However, the most distinct spectral differences are observed at 1320–900 cm^{-1} range. The 1800–1500 cm^{-1} range contains mainly amide I (1660 cm^{-1}) and amide II (centered at 1540 cm^{-1}) bands. Based on the literature, the positions of the amide I and II bands are indicative of protein structure [27,28]. There is no significant protein structural change among normal, adenoma and cancer tissues since the positions of the two bands do not change dramatically.

Fig. 3(b) shows the expanded spectra in the 1320–900 cm^{-1} region where the main spectral features are the stretching vibrations of PO_2^- from the nucleic acids and the C–O stretching vibrations from the carbohydrate residues in the glycogen (or collagen) [29,30]. In this case, the observable spectral features resulted mainly from the glycogen constituents rather than the nucleic acid components since the strong asymmetric vibration peak of the phosphate group at 1226 cm^{-1} were not observed. If the 1226 cm^{-1} band had been apparent, it could have been interpreted as the presence of nucleic acid. The C–O stretching absorptions from glycogen are featured at 1125, 1080 and 1040 cm^{-1} bands (marked as asterisks in Fig. 3(b)) and this assignment is analogous to that of Wong et al. [31]. Therefore, the most noticeable distinction of cancer and adenoma tissues compared to normal tissue samples in this region was the relatively lower band intensity (representing, therefore, lower content) of glycogen. Hence, for the discrimination between normal and malignant tissues, the spectral differences in this range can be effectively utilized.

The 1200–1140 cm^{-1} region is expanded in Fig. 3(c). The 1164 cm^{-1} band is the C–O band of the acyl chains of the membrane lipids. The position of this band has been shown to shift to higher wavenumber in cases where colon cancer has been diagnosed [31,32]. We have observed the same trend in stomach tissues. This band was shifted to 1172 cm^{-1} for both adenoma and cancer tissue samples.

Overall, IR spectral differences can used to discriminate malignant and normal stomach tissues. The biological origin of the spectral differences is difficult to fully address at present. However, these spectral differences can provide information that is selective for the discrimination among normal, adenoma and cancer, especially if a multivariate chemometric scheme is used in combination with the visual differences.

3.2. PCA analysis

Principal component analysis (PCA) [25,26] was used to investigate the extent to which IR features can differentiate normal, adenoma and cancer tissues. PCA is a widely accepted method that reduces complex spectral data into much fewer dimensions using principal components (also known as eigenvectors) and scores. Before performing PCA, all of the IR spectra were baseline corrected and normalized as described in the previous section. The first fifteen scores were calculated. With the use of eight or nine principal components (PCs), the total percent variance (TPV) was found to exceed 98%. TPV is an indicator of how much variation is aggregately represented by the calculated PCs. The scores represent the variation in the spectral data set, while eigenvalues are the relative weights of each individual PC. By summing eigenvalues and representing

the sum as a percentage, it can be estimated how much variance is described by the PCs.

Score plots (first score versus second score) using the 1800–900 cm^{-1} range are shown in Fig. 4. Plot (a) and (b) show the score scatter between normal and adenoma, and normal and cancer, respectively. Plot (c) shows the score scattering among normal, adenoma and cancer tissues. The discrimination between normal and malignant (adenoma and cancer) tissues is fairly clear since the difference in the spectral features could be visualized as shown in Fig. 3. The score clusters of adenoma and cancer are considerably overlapped with each other, which is consistent with the fact that the spectral features between adenoma and cancer are not visually distinct from each other. Therefore, it is reasonable to expect that the discrimination accuracy for adenoma or cancer would be diminished.

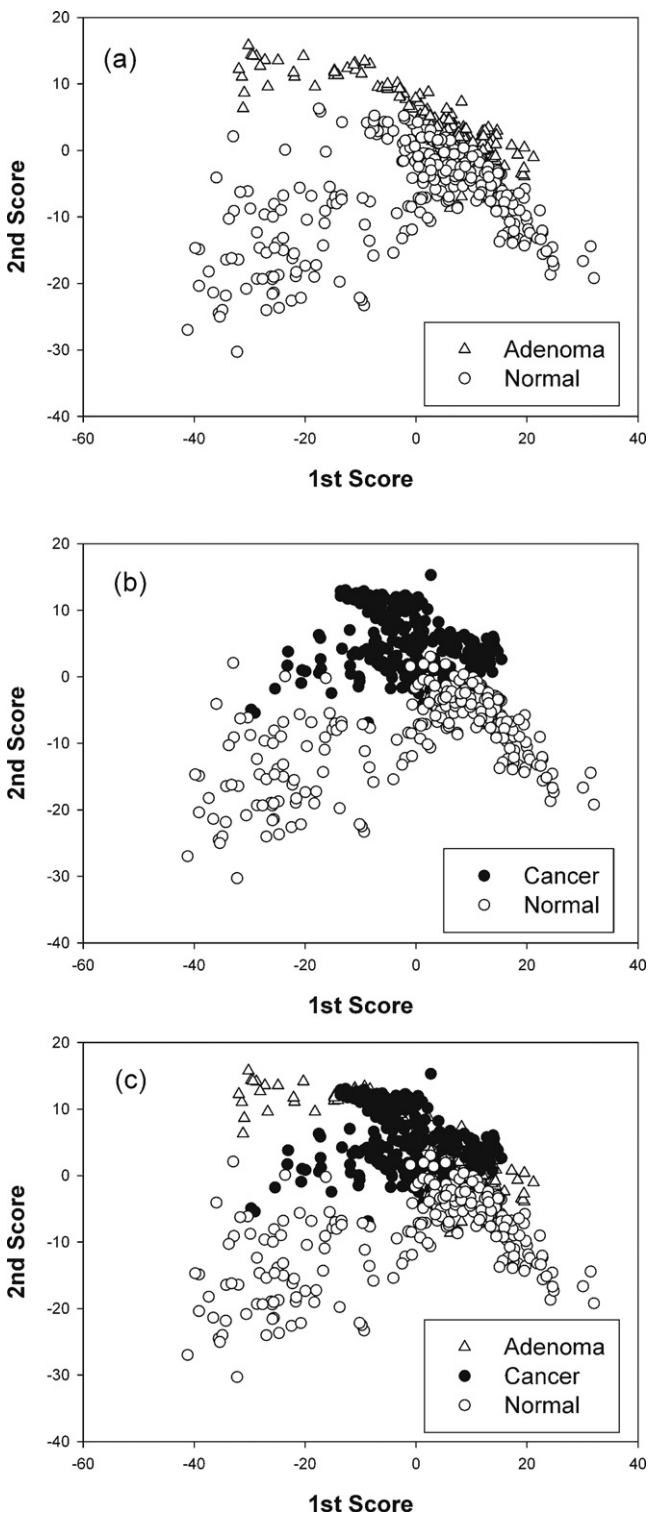

Fig. 4. The score plots showing the discrimination between normal and adenoma tissue (a), normal and cancer (b), and normal, adenoma and cancer tissues (c).

Soft independent modeling of class analogies (SIMCA) [25,26] was used to evaluate the discrimination accuracy quantitatively. To obtain optimal calibration performance, it is required to carefully select the proper spectral range(s) where informative spectral differences are available. When spectral features are clearly different, the selection of the spectral range(s) might not be a critical issue. However, in this study, it was necessary to find the spectral region best-suited for discrimination. It is especially important in this case because conclusive distinction is more important when methodologies such as this are applied to real medical diagnoses. In this case, we used four different spectral ranges (1800–900, 1800–1484, 1484–1186 and 1186–900 cm^{-1}). The first range used the whole spectral range and the second incorporated the protein information mostly where amide I and II bands are dominant. The third and fourth ranges focused on information from the nucleic acid and glycogen components, respectively.

When models are created, each class of materials is considered separately in SIMCA. For each class such as normal and adenoma, PCA is independently performed and the corresponding models (so-called disjointed class models) are developed individually. The following equations mathematically describe the more detailed procedure of SIMCA:

$$s_0^K = \left[\sum_{j=1}^{m}\sum_{i=1}^{n_K}\frac{(e_{ij}^K)^2}{(n_K - p_K - 1)(m - p_K)}\right]^{1/2} \quad (1)$$

$$s_u^K = \left[\sum_{j=1}^{m}\frac{(e_{uj}^K)^2}{(m - p_K)}\right]^{1/2} \quad (2)$$

$$F = \frac{(s_u^K)^2}{(s_0^K)^2} \quad (3)$$

$$(s_{lim}^K) = (s_0^K)F^{0.05} \quad (4)$$

In Eq. (1), s_0^K is the standard deviation of the training set for class K, n_k is the number of objects, p_k is the number of significant principal components in class K, m is the number of variables, and e_{uj} is the residual. The standard deviation of the unknown spectrum, s_u^K, is then calculated using e_{uj} value. In SIMCA, an unknown spectrum is identified as a specific group (or other group) by comparing the F-distribution with the 95% confidence interval. If the F-value is larger than the critical F-value at a given level of significance, one can conclude that the distance (u) from class K is significantly larger than that of the

Table 2
The assigned number of spectra and the corresponding subjects for calibration and validation set

	Calibration set	Validation set
Normal	248 (subject 1, 2, 7–10)	100 (subject 1, 3)
Adenoma	166 (subject 2, 9, 10)	10 (subject 1)
Cancer	296 (subject 4, 6–8, 11)	106 (subject 3, 5)

class K as a whole and it is determined to be excluded from class K. If the distance is not significantly larger than F-value, the sample is determined to be a member of class K.

To perform SIMCA, each dataset was divided to create a calibration set and a validation set. The calibration set was used to build the SIMCA model for each case (normal, adenoma and cancer) and each developed model was validated using the validation set. The divisions of the data set are described in Table 2.

The best result was achieved using the whole spectral range (1800–900 cm^{-1}). This result indirectly supports the fact that the chemical differences among normal, adenoma and cancer tissues do not originate solely from one specific alteration, such as protein structure change. Other components are also ostensibly a factor in the differences detected in the spectral patterns of the tissue types even though these other contributions are relatively small. The resulting SIMCA prediction accuracies for normal, adenoma and cancer were 77%, 30% and 87%, respectively. The numbers of principal components used in each model were 8, 9 and 8, respectively. The prediction accuracy for cancer is the best and that for normal is quite acceptable. However, the prediction accuracy for adenoma is not satisfactory due to its similar spectral features with that of cancer. Additionally, more adenoma samples would allow better statistical estimation of diagnosis (only 10 spectra from one subject were predicted in this study).

4. Conclusion

The results demonstrate that IR micro-spectroscopy is feasible as a tool for fast (a few minutes when an IR array detector is used) and simple diagnosis for stomach cancer, and possibly also for adenoma even though its prediction accuracy for the latter is not as efficient. The differentiation between normal and malignant (cancer and adenoma) stomach tissue would be possible based on the score plot. However, a greater number of subjects and samples are needed to ensure the utility of the proposed method for real clinical diagnoses. We are currently expanding the IR spectral database.

Although factor based analyses such as PCA and SIMCA are frequently employed to differentiate normal and malignant tissues, there is still a need to improve the discrimination capability for samples such as those used in this study. This is because the PCA scores of the adenoma and cancer samples were significantly overlapped. One possible way to improve the discrimination in a case such as this is to acquire more selective and sensitive IR spectra. However, this is practically limited since a high energy IR source such as a synchrotron would have to be used. Another possible solution for this predicament would be to develop new chemometric algorithms that can more effectively extract relevant information and improve discrimination capability. This is the more practical approach to avoid the burden of large investments in instrumentation. Chemometric algorithms that can improve discrimination will be necessary for more accurate biomedical diagnosis using spectroscopy. We are currently developing new chemometric methods that can diagnose spectral differences more efficiently and improve discrimination between stomach adenoma and cancer.

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